



Amazon Sales & Operations Analytics Dashboard

I. Project Overview

This project is a complete Business Intelligence solution built using Power BI to analyze Amazon sales performance, customer behavior, product profitability, and shipping operations.

The dashboard was designed to help management make data-driven decisions by transforming raw transactional data into meaningful business insights through interactive visualizations, KPI tracking, and operational analytics.

The solution includes advanced data cleaning, data modeling, custom DAX measures, calendar table creation, and multi-page dashboard reporting.

II. Project Objective

The main objective of this project was to analyze business performance across multiple dimensions including sales, customers, products, and shipping operations.

The dashboard helps identify:

- Revenue and profit trends
- High-performing and loss-making products
- Valuable customer segments
- Customer retention patterns
- Delivery efficiency and operational bottlenecks
- Impact of discounts on profitability

The goal was to create an executive-level interactive dashboard that supports business decision-making and operational optimization.

III. Business Problems Solved

This dashboard addresses several key business problems:

- Difficulty in tracking overall business performance
- Lack of visibility into product profitability
- Identifying products generating high sales but low profit
- Understanding customer purchasing behavior and retention
- Analyzing the impact of discounts on revenue and margins
- Monitoring shipping efficiency and delayed deliveries
- Identifying operational bottlenecks across regions and ship modes

The dashboard transforms raw sales data into actionable insights for strategic decision-making.

IV. Tools & Technologies Used

- Power BI – Dashboard Development & Visualization
- Power Query – Data Cleaning & Transformation
- DAX – KPI Calculations & Business Logic
- Data Modeling – Relationship Building
- Custom Calendar Table – Time Intelligence Analysis
- Excel/CSV Dataset – Data Source

V. Data Preparation & Transformation

The dataset was cleaned and transformed using Power Query to improve data quality and analytical performance.

Data preparation steps included:

- Handling null and missing values
- Changing incorrect data types
- Creating calculated columns

- Building a custom calendar table
- Establishing relationships between tables
- Optimizing the data model for efficient reporting

VI. Dashboard Pages Overview

1. Executive Summary Dashboard



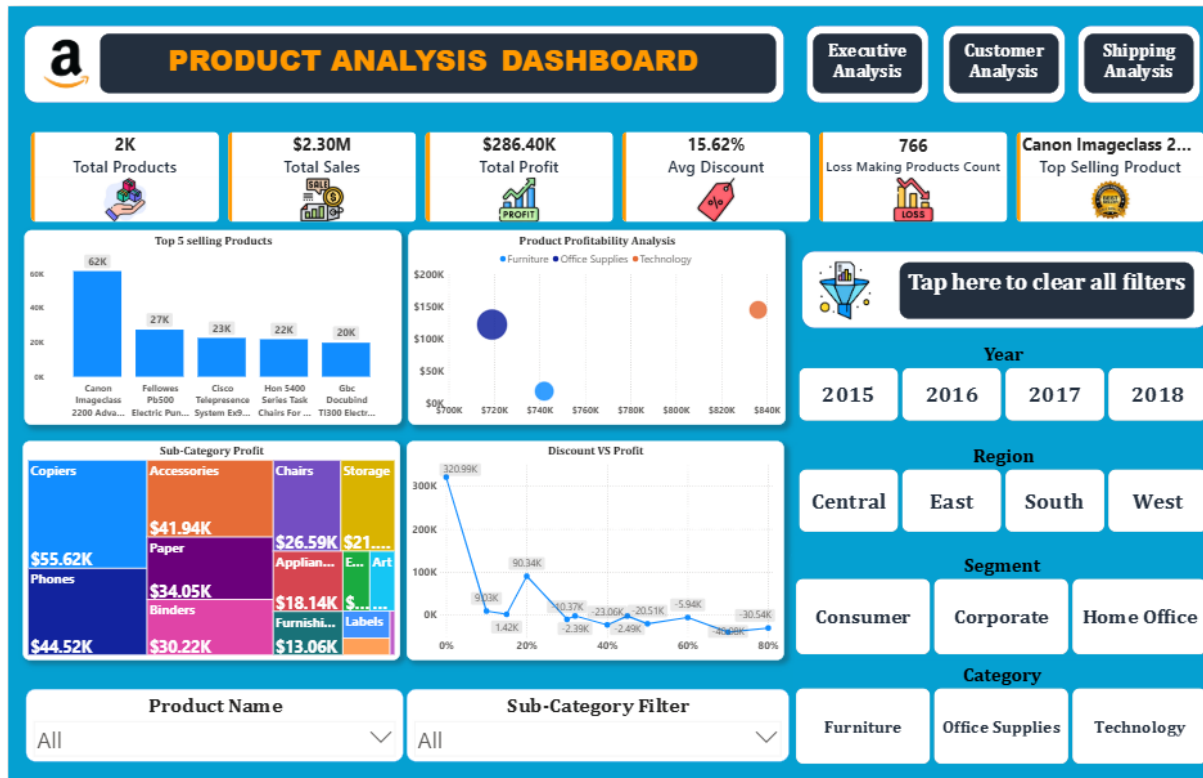
The Executive Summary dashboard provides a high-level overview of overall business performance using KPI cards and interactive visualizations.

Key insights include:

- Total Sales, Profit, Orders, Quantity, Profit Margin percent and Average Shipping Days.
- Sales and Profit trends over time.
- Regional sales distribution

- Category-wise performance analysis
- Monthly profitability patterns

2. Product Analysis Dashboard

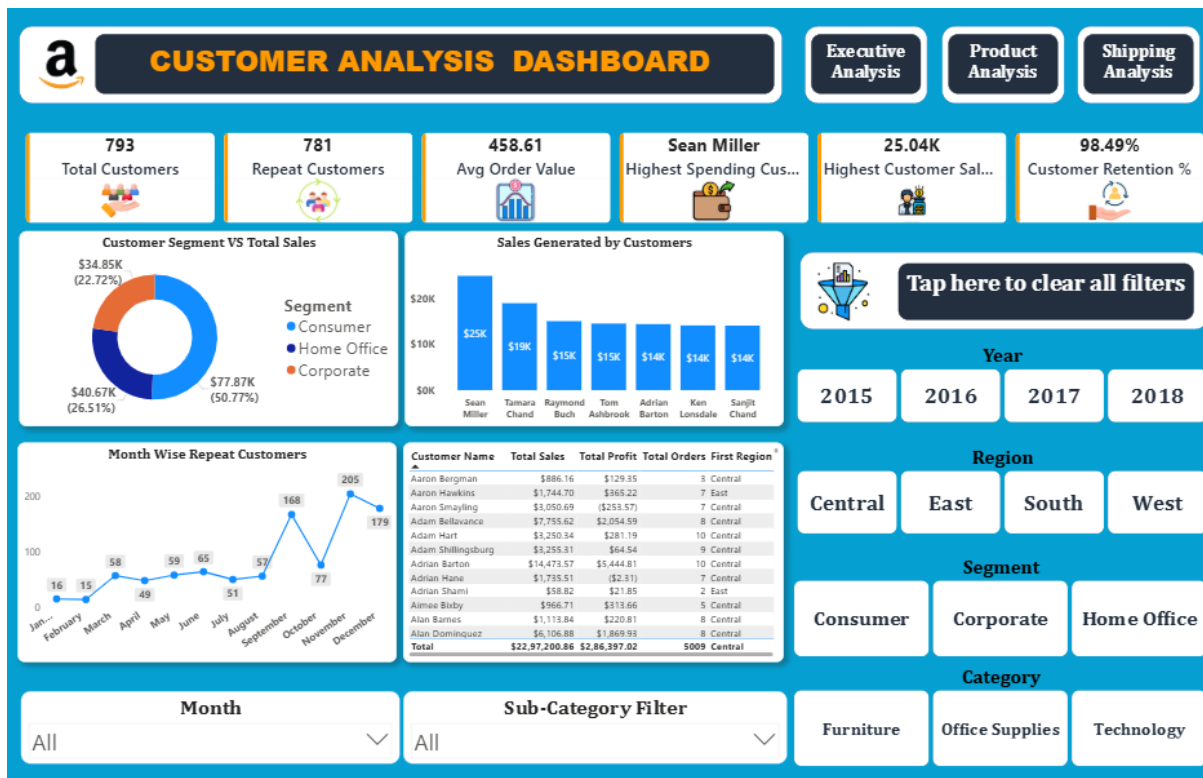


The Product Analysis dashboard focuses on product-level profitability and sales performance.

Key insights include:

- Total Products, Sales, Profit, Average Discount, Loss Making Products Count and Top Selling Products.
- Top-selling products
- Product profitability analysis
- Loss-making products identification
- Sub-category profit contribution
- Discount impact on profitability
- Product performance comparison across categories

3. Customer Analysis Dashboard

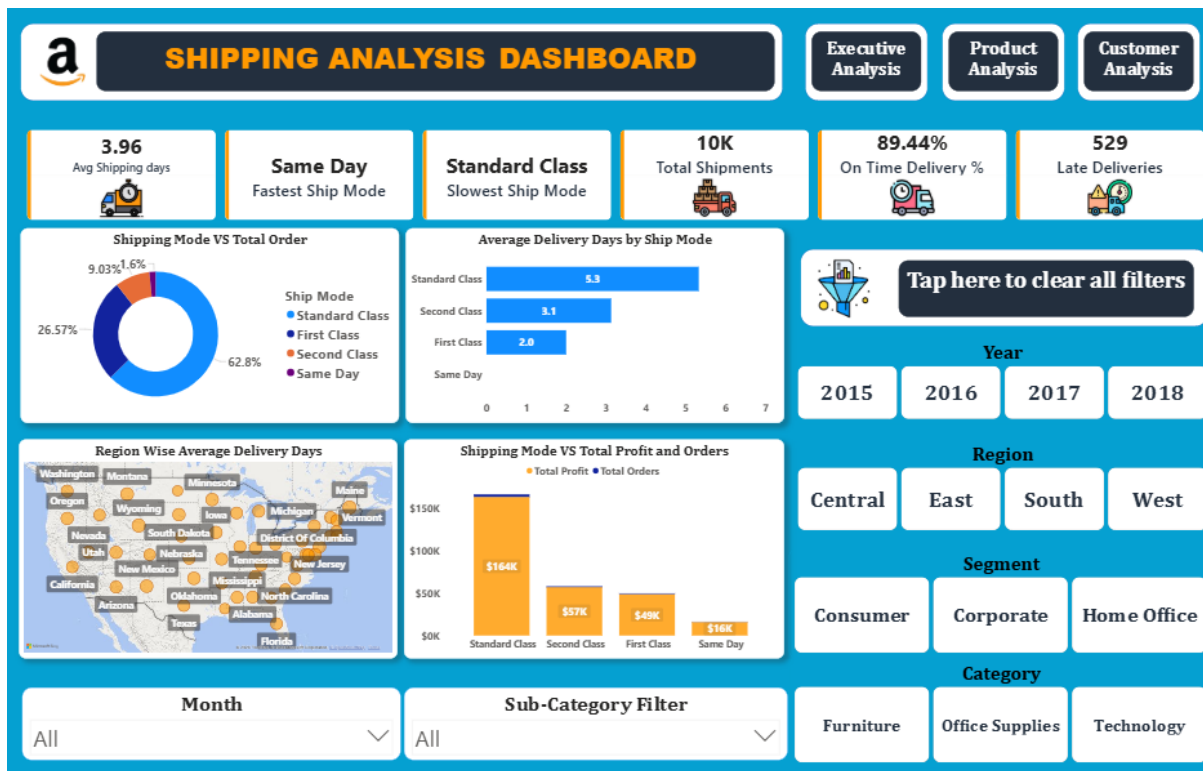


The Customer Analysis dashboard helps analyze customer behavior, customer contribution, and retention patterns.

Key insights include:

- Total Customers, Repeat Customers, Average Order Value, Highest Spending Customer, Highest Customer Sales and Customer Retention Percentage.
- Customer segmentation analysis
- Highest spending customers
- Repeat customer trends
- Revenue contribution by customers
- Customer retention percentage
- Regional customer distribution

4. Shipping & Operations Dashboard



The Shipping & Operations dashboard evaluates delivery performance and operational efficiency.

Key insights include:

- Average Shipping days, Fastest Shipping Mode, Slowest Shipping Mode, Total Shipments, On-time Delivery Percent and Late Deliveries.
- Average shipping days
- Fastest and slowest shipping modes
- On-time delivery percentage
- Late delivery analysis
- Regional delivery efficiency
- Shipping mode profitability impact

VII. Key KPIs Created Using DAX

- Total Sales
- Total Profit
- Total Orders
- Profit Margin %
- Average Shipping Days
- On-Time Delivery %
- Customer Retention %
- Repeat Customers
- Highest Spending Customer
- Loss Making Products Count
- Top Selling Product.

VIII. Key Business Insights

- Technology category generated the highest sales contribution
- Certain products showed high sales but low profitability
- Heavy discounts negatively affected profit margins
- Consumer segment contributed the highest revenue share
- Standard Class shipping had the longest average delivery time
- Some regions experienced operational delivery delays
- Customer retention rate remained significantly high

IX. Challenges Faced

- Handling duplicate order records
- Creating optimized DAX measures
- Building custom time intelligence calculations
- Designing interactive multi-page dashboards
- Implementing operational delivery logic without required delivery dates
- Maintaining dashboard performance and visual consistency

X. Conclusion

This project demonstrates how Business Intelligence tools can transform raw business data into actionable insights.

The dashboard enables management to monitor sales performance, optimize operations, improve customer understanding, and make strategic business decisions through interactive reporting and data-driven analysis.

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